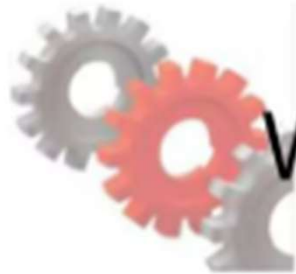


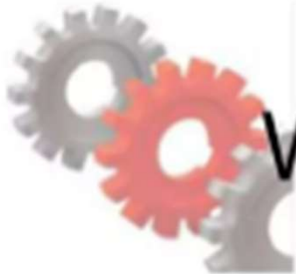
IPE 4111

Supply Chain Management



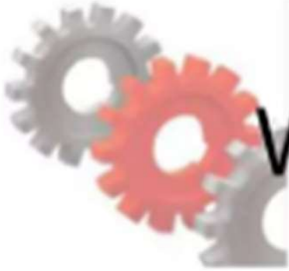
What is a Supply Chain?

- All stages involved, directly or indirectly, in fulfilling a customer request
- Includes manufacturers, suppliers, transporters, warehouses, retailers, and customers
- Within each company, the supply chain includes all functions involved in fulfilling a customer request (product development, marketing, operations, distribution, finance, customer service)



What is a Supply Chain?

- Customer is an integral part of the supply chain
- Includes movement of products from suppliers to manufacturers to distributors and information, funds, and products in both directions
- May be more accurate to use the term “supply network” or “supply web”
- Typical supply chain stages: customers, retailers, distributors, manufacturers, suppliers
- All stages may not be present in all supply chains (e.g., no retailer or distributor for Dell)



What is a Supply Chain?

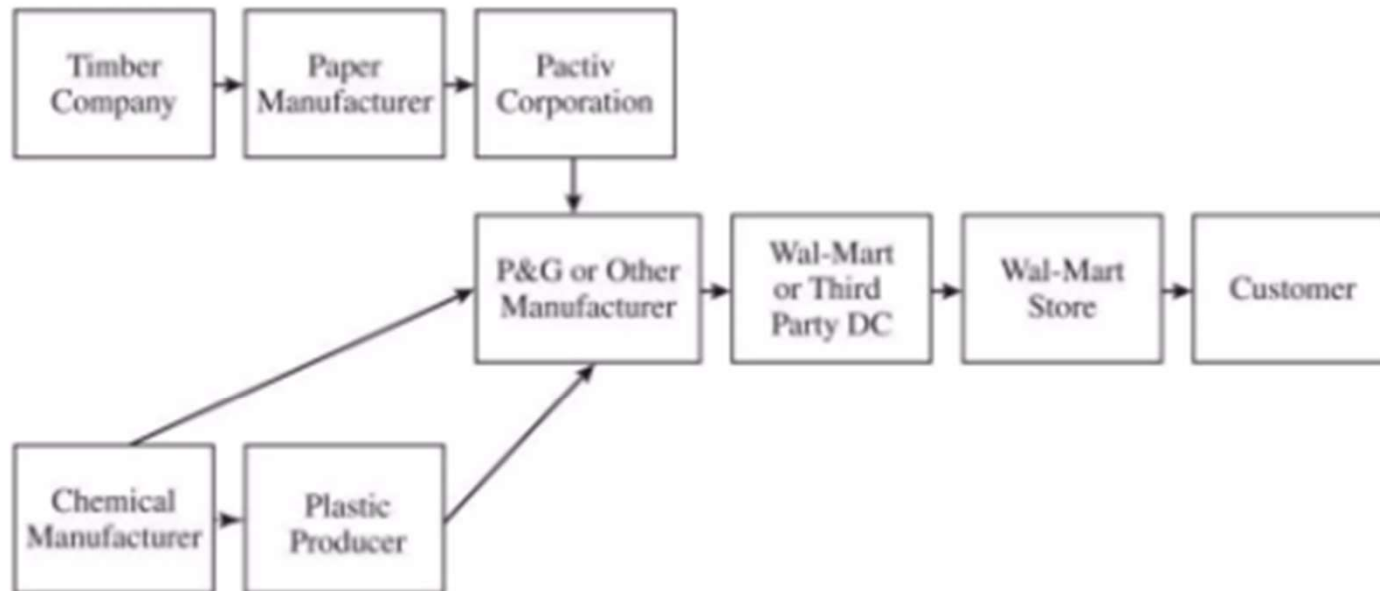


Figure 1-1

Flows in a Supply Chain

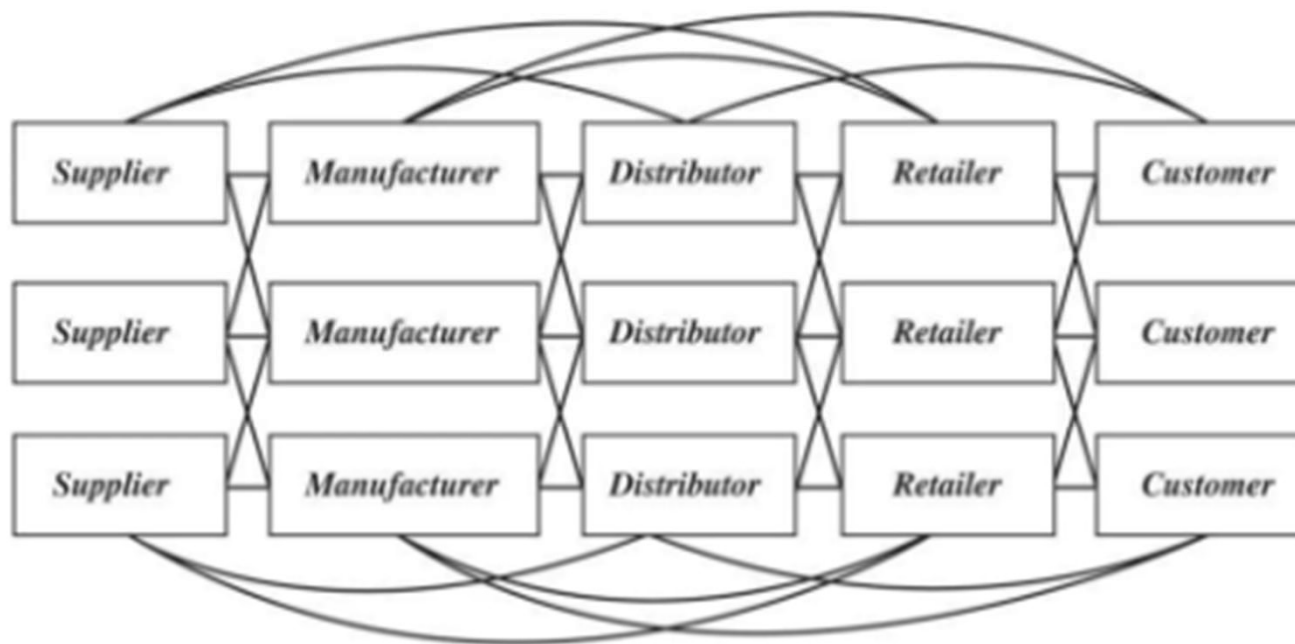


Figure 1-2



The Objective of a Supply Chain

- Maximize overall value created

Supply Chain Surplus

= Customer Value – Supply Chain Cost



The Objective of a Supply Chain

- Example: a customer purchases a wireless router from Best Buy for \$60 (revenue)
- Supply chain incurs costs (information, storage, transportation, components, assembly, etc.)
- Difference between \$60 and the sum of all of these costs is the supply chain profit
- Supply chain profitability is total profit to be shared across all stages of the supply chain
- Success should be measured by total supply chain profitability, not profits at an individual stage



The Objective of a Supply Chain

- Customer the only source of revenue
- Sources of cost include flows of information, products, or funds between stages of the supply chain
- Effective supply chain management is the management of flows between and among supply chain stages to maximize total supply chain surplus



Decision Phases of a Supply Chain

- Supply chain strategy or design
 - How to structure the supply chain over the next several years
- Supply chain planning
 - Decisions over the next quarter or year
- Supply chain operation
 - Daily or weekly operational decisions



Supply Chain Strategy or Design

- Decisions about the structure of the supply chain and what processes each stage will perform
- Strategic supply chain decisions
 - Locations and capacities of facilities
 - Products to be made or stored at various locations
 - Modes of transportation
 - Information systems
- Supply chain design must support strategic objectives
- Supply chain design decisions are long-term and expensive to reverse – must take into account market uncertainty



Supply Chain Planning

- Planning decisions:
 - Which markets will be supplied from which locations
 - Planned buildup of inventories
 - Subcontracting, backup locations
 - Inventory policies
 - Timing and size of market promotions
- Must consider in planning decisions demand uncertainty, exchange rates, competition over the time horizon



Supply Chain Operation

- Time horizon is weekly or daily
- Decisions regarding individual customer orders
- Supply chain configuration is fixed and operating policies are determined
- Goal is to implement the operating policies as effectively as possible
- Allocate orders to inventory or production, set order due dates, generate pick lists at a warehouse, allocate an order to a particular shipment, set delivery schedules, place replenishment orders
- Much less uncertainty (short time horizon)



Process View of a Supply Chain

- *Cycle View*: processes in a supply chain are divided into a series of cycles, each performed at the interfaces between two successive supply chain stages
- *Push/Pull View*: processes in a supply chain are divided into two categories depending on whether they are executed in response to a customer order (*pull*) or in anticipation of a customer order (*push*)



Cycle View of Supply Chain Processes

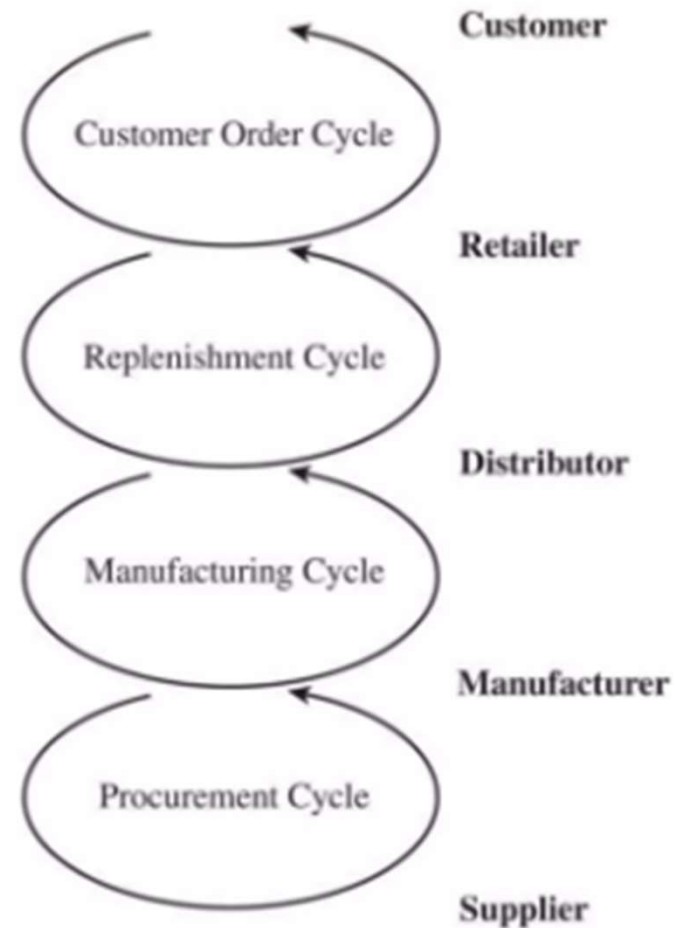


Figure 1-3



Cycle View of Supply Chain Processes

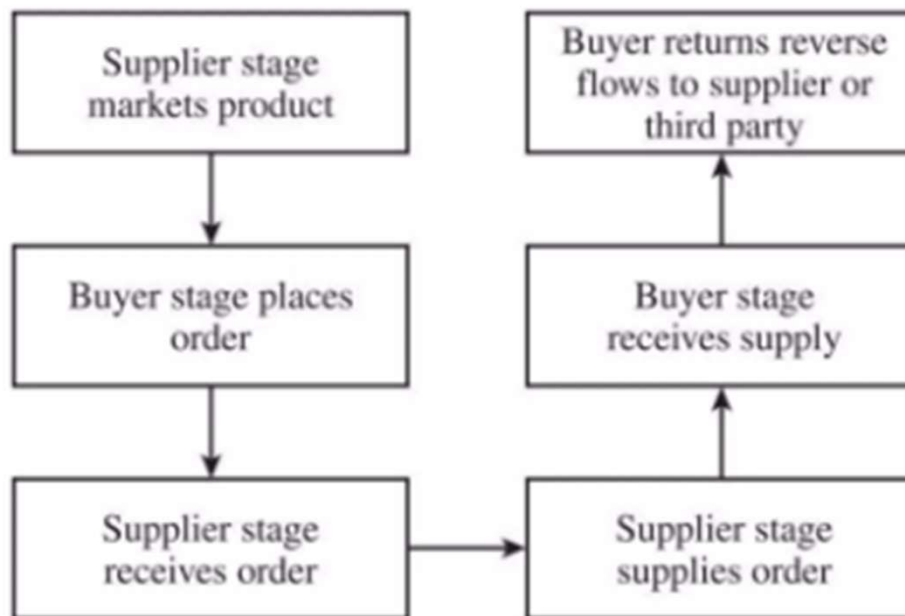


Figure 1-4



Push/Pull View of Supply Chains

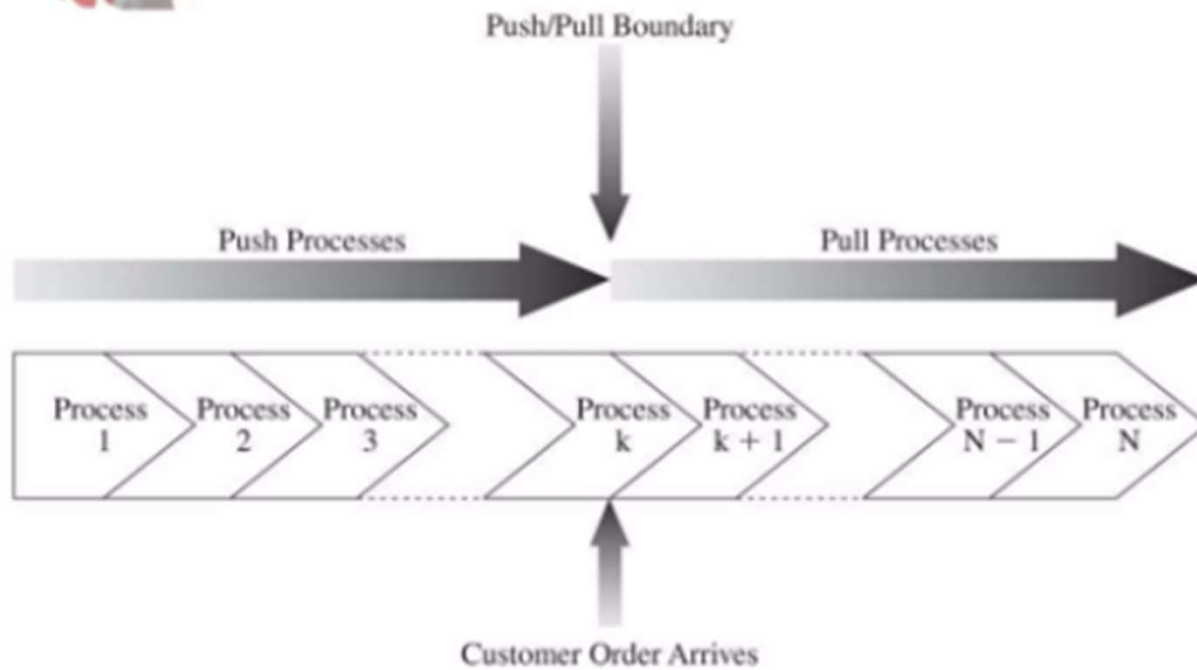


Figure 1-5



Push/Pull View of Supply Chain Processes

- Supply chain processes fall into one of two categories depending on the timing of their execution relative to customer demand
- Pull: execution is initiated in response to a customer order (reactive)
- Push: execution is initiated in anticipation of customer orders (speculative)
- Push/pull boundary separates push processes from pull processes



Push/Pull View of Supply Chain Processes

- Useful in considering strategic decisions relating to supply chain design – more global view of how supply chain processes relate to customer orders
- Can combine the push/pull and cycle views
 - L.L. Bean
 - Dell
- The relative proportion of push and pull processes can have an impact on supply chain performance



Push/Pull View of – L.L. Bean

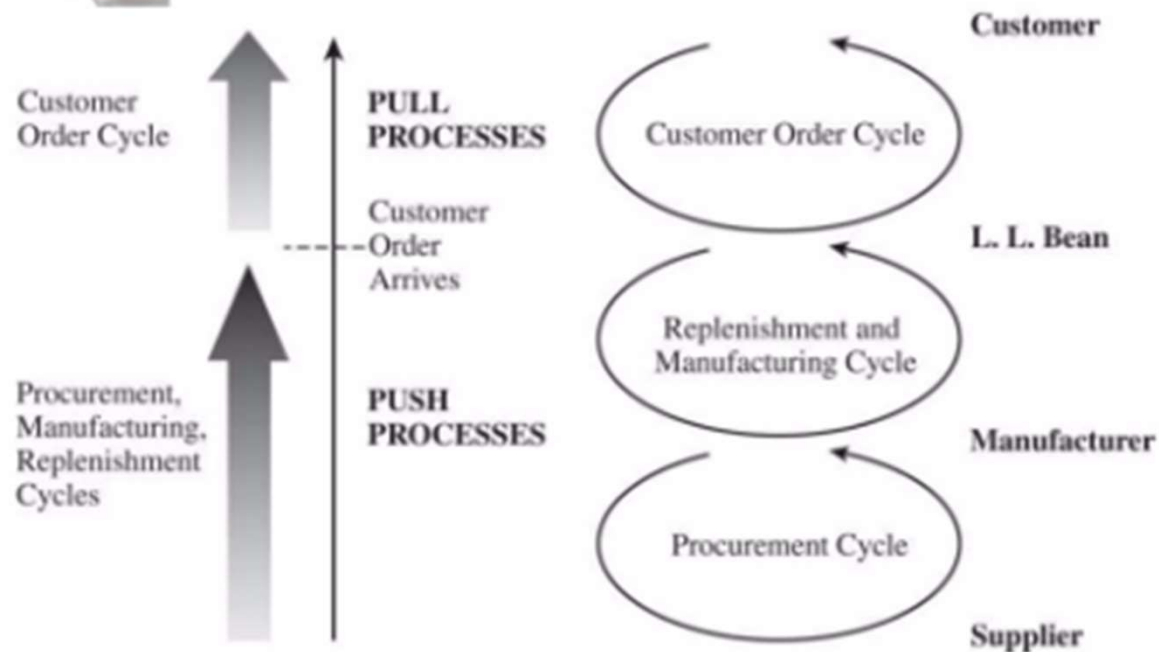


Figure 1-6



Competitive and Supply Chain Strategies

- Competitive strategy defines the set of customer needs a firm seeks to satisfy through its products and services
- Product development strategy specifies the portfolio of new products that the company will try to develop
- Marketing and sales strategy specifies how the market will be segmented and product positioned, priced, and promoted
- Supply chain strategy determines the nature of material procurement, transportation of materials, manufacture of product or creation of service, distribution of product
- All functional strategies must support one another and the competitive strategy

The Value Chain



Figure 2-1



Achieving Strategic Fit

- Strategic fit – competitive and supply chain strategies have aligned goals
- A company may fail because of a lack of strategic fit or because its processes and resources do not provide the capabilities to execute the desired strategy



Achieving Strategic Fit

1. The competitive strategy and all functional strategies must fit together to form a coordinated overall strategy.
2. The different functions in a company must appropriately structure their processes and resources to be able to execute these strategies successfully.
3. The design of the overall supply chain and the role of each stage must be aligned to support the supply chain strategy.



Step 1: Understanding the Customer and Supply Chain Uncertainty

- Quantity of product needed in each lot
- Response time customers will tolerate
- Variety of products needed
- Service level required
- Price of the product
- Desired rate of innovation in the product



Step 1: Understanding the Customer and Supply Chain Uncertainty

- *Demand uncertainty* – uncertainty of customer demand for a product
- *Implied demand uncertainty* – resulting uncertainty for the supply chain given the portion of the demand the supply chain must handle and attributes the customer desires



Step 2: Understanding Supply Chain Capabilities

- How does the firm best meet demand?
- Supply chain responsiveness is the ability to
 - Respond to wide ranges of quantities demanded
 - Meet short lead times
 - Handle a large variety of products
 - Build highly innovative products
 - Meet a very high service level



Step 2: Understanding Supply Chain Capabilities

- Responsiveness comes at a cost
- Supply chain efficiency is the inverse to the cost of making and delivering the product to the customer
- The *cost-responsiveness efficient frontier* curve shows the lowest possible cost for a given level of responsiveness

Cost-Responsiveness Efficient Frontier

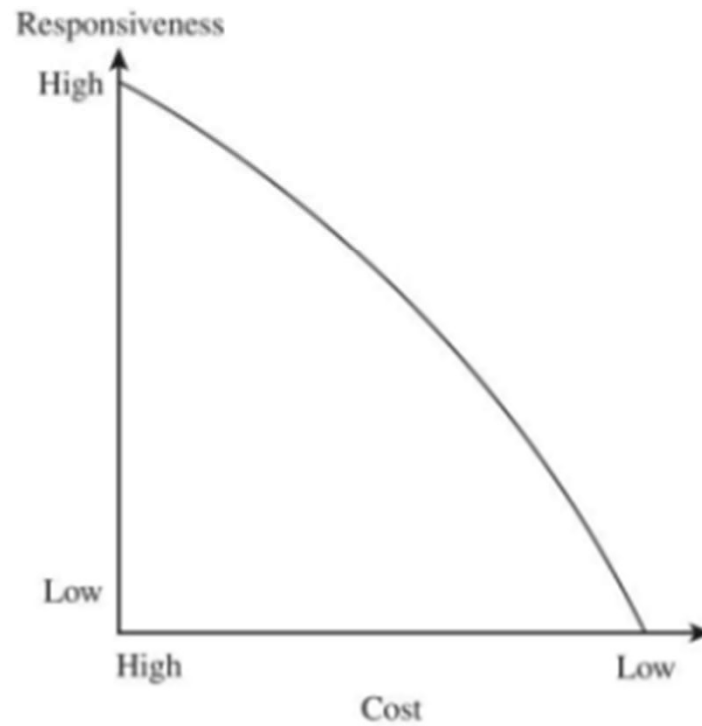


Figure 2-3

Responsiveness Spectrum

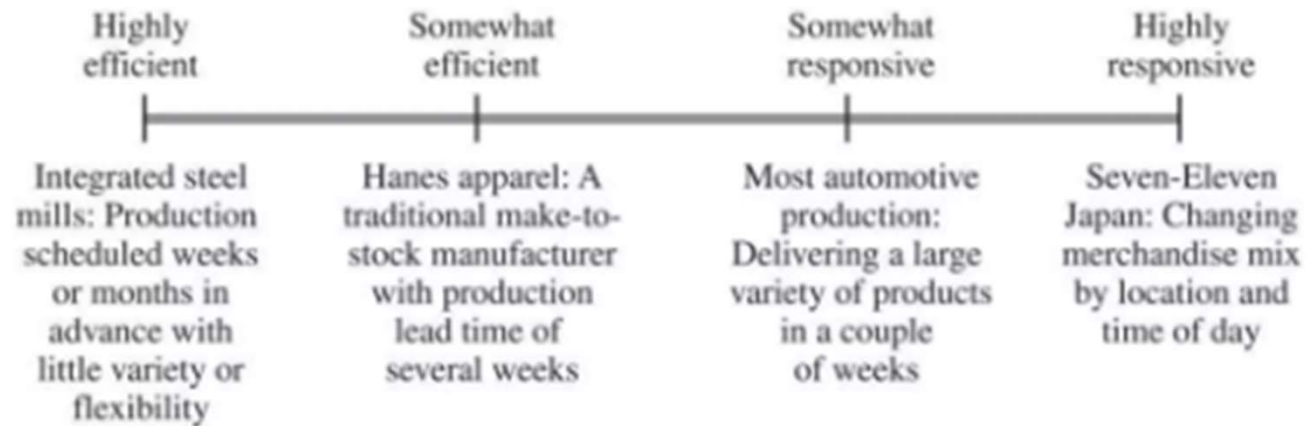


Figure 2-4



Step 3: Achieving Strategic Fit

- Ensure that the degree of supply chain responsiveness is consistent with the implied uncertainty
- Assign roles to different stages of the supply chain that ensure the appropriate level of responsiveness
- Ensure that all functions maintain consistent strategies that support the competitive strategy

Zone of Strategic Fit

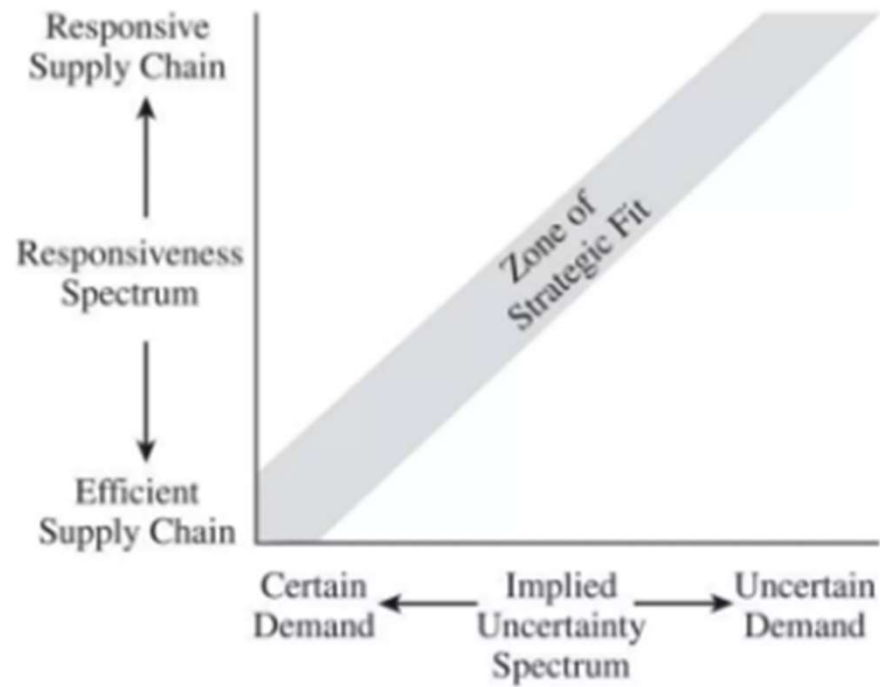


Figure 2-5